

Then create a substring that checks the number of people being detected. When the program is run, it calls TensorFlow modules for detection

and monitoring. It will estimate the number of people and display images of the persons present in that area on your computer screen.

```
import os
import cv2
import numpy as np
from picamera.array import PiRGBArray
from picamera import PiCamera
import tensorflow as tf
import argparse
import sys
from gpiozero import LED
led = LED(17)
import time

a=time.time()
IM_WIDTH = 880
IM_HEIGHT = 880
#IM_WIDTH = 640
#IM_HEIGHT = 480
#IM_WIDTH = 640    Use smaller resolution for
#IM_HEIGHT = 480  slightly faster framerate

|

# Select camera type (if user enters --usbcam when calling this script,
# a USB webcam will be used)
camera_type = 'picamera'
parser = argparse.ArgumentParser()
parser.add_argument('--usbcam', help='Use a USB webcam instead of picamera',
                    action='store_true')
args = parser.parse_args()
if args.usbcam:
    camera_type = 'usb'
```

Fig. 11: Required libraries in the code for crowd detection

```
# Grab path to current working directory
CWD_PATH = os.getcwd()

# Path to frozen detection graph .pb file, which contains the model that is used
# for object detection.
PATH_TO_CKPT = os.path.join(CWD_PATH,MODEL_NAME,'frozen_inference_graph.pb')

# Path to label map file
PATH_TO_LABELS = os.path.join(CWD_PATH,'data','crowd.pbtxt')

# Number of classes the object detector can identify
NUM_CLASSES = 4
```

Fig. 12: Crowd .pbtxt defined in the code

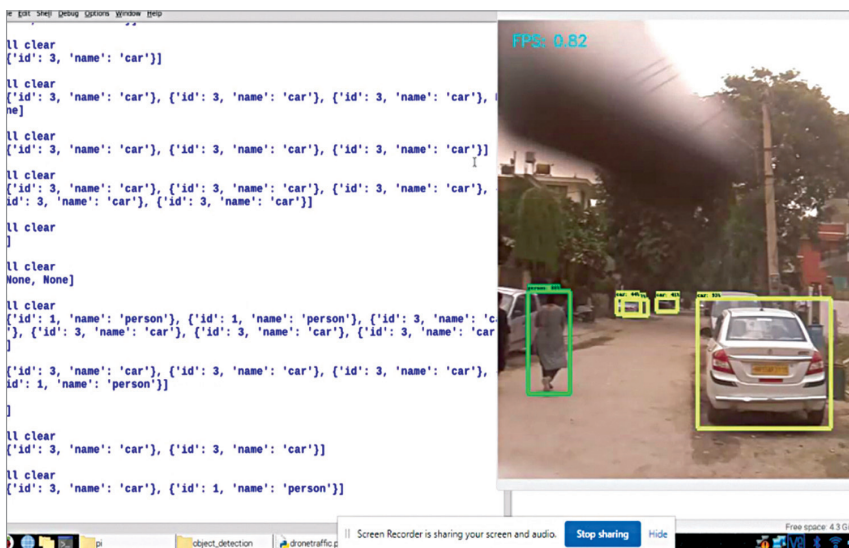


Fig. 13: Traffic detection on a street

EFY Note

The source code of this project is available for free download at source.efymag.com

Traffic detection

The same drone can be used for monitoring vehicular traffic on the streets of a city or town (refer Fig. 13). For this, you must create traffic.pbtxt file in the project folder and add the list of objects, or display_names like car, bus, people, signboard, truck, etc in the file.

When the dronetraffic.py program is run, it will first check traffic.pbtxt file to start detecting the number of cars, buses, bikes, etc on the road. The program saves the images of the cars and other details in a text file in a folder. If it detects too many vehicles or people on the road/street, it will alert the authority by displaying messages such as “Heavy traffic” in case of vehicles, or “Too many people” in case of commuters on the road.

The project can be extended for use in forest survey and animal detection using the same code but with a different .pbtxt file. For animal detection, create a .pbtxt file with list of the animals to be detected and then change the substring in the code to output the results on computer screen.

Testing

You need to run each code (dronecode1.py, dronecrowd.py, and dronetraffic.py) separately at different times for different applications. For example, when the dronecode1.py is up and running, mount it on the drone. You can see live images being captured by the camera on your computer screen. You can send the drone with camera to a designated area for detection and monitoring. The distance the drone can fly depends on the Wi-Fi range. **EFY**

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